Plant Disease Detection Using Deep Learning

A Deep Learning Approach to Classify Leaf Disease from Image

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***Abstract*— Plant diseases pose a significant threat to agriculture and global food security. Accurate and early identification of plant diseases plays a vital role in minimizing crop damage and ensuring food security. Traditional visual inspection is often slow and inefficient, especially across large and symptomatically complex farming regions. Therefore, the need for an automated and efficient plant disease detection system becomes critical. The work combines image processing, machine learning, and practical software development to create a scalable solution for smart agriculture. By automating plant disease detection, it reduces the dependency on manual diagnosis and enables timely intervention. This work proposes a deep learning-based approach using a Convolutional Neural Network (CNN) to automatically detect and classify plant leaf diseases. The model architecture includes three convolutional layers followed by max-pooling, a dense hidden layer, and a softmax output layer for multi-class classification. A finely tuned Convolutional Neural Network (CNN) trained on the PlantDisease dataset was used, achieving an impressive 98.97% classification accuracy. The model design was intentionally kept simple by relying solely on CNN architecture, avoiding additional complex optimization techniques. This high accuracy suggests that CNN could support more scalable and efficient plant disease monitoring, which would eventually enable improved farming methods.**

***Index Terms*—CNN, Deep Learning, Classify Leaf Disease.**

I. INTRODUCTION

Ensuring food availability and supporting millions of lives, agriculture continues to be a key driver of economic and societal progress. A production is a very old means of obtaining food. It is a vital source of income for people all around the world. However, one of the most persistent challenges faced by farmers is the outbreak of plant diseases, which can lead to significant reductions in crop yield and quality. Too many pesticides are harmful to crops and farmland. Getting expert advice will help you avoid misusing chemicals on plants. Plants have been the focus of many researchers to aid farmers and others involved in agriculture. These diseases can be caused by various pathogens, such as fungi, bacteria, and viruses, or by non-infectious factors like unfavorable environmental conditions. Farmers and agricultural experts manually examine plants, looking for visible symptoms like discoloration, spots, and lesions. While effective to some degree, this approach has limitations in terms of speed, accuracy, and scalability

Agriculture stands as a vital pillar of the world's economy, playing a central role in ensuring food availability, creating job opportunities, and fostering growth in rural regions. However, one of the greatest challenges farmers face today is the constant threat of plant diseases, which significantly affect crop health and agricultural productivity. Plant diseases, if left undetected or untreated, can result in extensive crop loss, reduced yield quality, and increased production costs. Manual identification and treatment of plant diseases require agricultural expertise and are time-consuming, labor-intensive, and often impractical in large-scale farming environments. Plant diseases can be effectively reduced by using deep learning algorithms like CNN, which enable accurate and automated detection of infected plants, allowing for timely intervention and improved plant health management. The use of Convolutional Neural Networks (CNN) in plant disease detection offers significant advantages for farmers by enabling early, accurate, and automated identification of crop diseases. This helps farmers avoid large-scale crop damage and improves yield quality and quantity. By detecting diseases like Black Rot, ESCA, and Leaf Blight efficiently, CNNs help farmers make timely, Ultimately, the adoption of CNN-based plant disease detection empowers farmers with precision farming tools that enhance productivity, profitability, and food security. Convolutional Neural Networks (CNNs) are highly effective in this domain, This project focuses on developing a CNN-based deep learning model to automatically detect and classify plant diseases using images of plant leaves. The core objective is to create an end-to-end pipeline that can preprocess image data, train a robust CNN model, evaluate its performance, and provide real-time predictions for new input images.

Abbreviations and Acronyms

CNN-Convolutional Neural Networks
RF-Random Forest
RNN-Recurrent Neural Networks
LSTM-Long Short Term Memory
ESCA- Eutypa Stereum, And Cryptovalsa-Associated Disease

II.LITERATURE SURVEY

1.Sladojevic, S., et al. (2016). Deep Neural Networks Based Recognition of Plant Diseases by Leaf Image Classification. Srdjan Sladojevic is a researcher known for his work in deep learning applied to agriculture. In 2016, he led a study that used convolutional neural networks (CNNs) to detect plant diseases from leaf images with high accuracy. He was affiliated with the University of Novi Sad, Serbia, during this research. His work showed how AI can support farmers in identifying plant diseases

quickly and effectively. This influential research laid the foundation for mobile apps and smart farming tools that use AI for plant health monitoring.

2.Mohanty, S. P., et al. (2016). Using Deep Learning for Image-Based Plant Disease Detection.

Sharada Prasanna Mohanty is a researcher best known for co-authoring the pioneering 2016 study that applied deep convolutional neural networks to identify plant diseases from leaf images with remarkable accuracy . she helped train a model on a dataset of over 54,000 images covering 14 crop species and 26 disease conditions, achieving a top test accuracy of 99.35 % .Her efforts drew attention to the potential of smartphone-assisted disease diagnosis, enabling scalable precision agriculture solutions. Mohanty's work demonstrated that deep learning models could operate rapidly on mobile devices, classifying images in less than a second. Her research has become a foundational reference in agricultural AI, inspiring further advances in field-deployable plant disease detection tools.

3.Amara, J., et al. (2017). A Deep Learning-Based Approach for Banana Leaf Diseases Classification.

Jihen Amara is a computer scientist and researcher who was affiliated with Friedrich Schiller University Jena at the time of the 2017 publication. She co-authored the pioneering paper "A Deep Learning–Based Approach for Banana Leaf Diseases Classification," in which she and her colleagues utilized the LeNet convolutional neural network to automatically detect and categorize banana leaf diseases. The model demonstrated impressive performance under challenging real-world imaging conditions varying light, complex backgrounds—achieving high accuracy (up to ~99.7%) in classifying leaf diseases. Amara's work represents an early and influential application of deep learning in precision agriculture, contributing significantly to the field of crop disease detection

4.Mazoyer M, Roudart L (2006) A history of world agriculture: from the neolithic age to the current crisis. NYU Press

Marcel Mazoyer is a renowned French agronomist and professor emeritus at AgroParisTech, known for his extensive work in agricultural development and history. He co-authored A History of World Agriculture: From the Neolithic Age to the Current Crisis with historian Laurence Roudart in 2006, offering a comprehensive analysis of global agricultural evolution. Laurence Roudart, also an expert in agronomy and food systems, has collaborated with Mazoyer on several works focusing on sustainable agriculture and food sovereignty. Together, they emphasize how historical agricultural systems shape current global inequalities and propose strategies for equitable and sustainable development.

5.Chowdhury MEH et al (2021) Automatic and reliable leaf disease detection using deep learning techniques. AgriEngineering 3(2):294–312

Muhammad E. H. Chowdhury is a researcher specializing in the application of deep learning and artificial intelligence to agriculture. In the 2021 paper "Automatic and Reliable Leaf Disease Detection Using Deep Learning Techniques", he and colleagues utilized EfficientNet and U-Net architectures on a dataset of 18,161 tomato leaf images yielding segmentation accuracies around 98% and classification accuracies exceeding 99% His work is notable for integrating both segmentation and classification in a unified framework, demonstrating superior performance compared to earlier CNN-based methods. This contribution highlights Chowdhury's role in advancing precision agriculture through robust, real-world AI solutions.

6.Sharma S, Kooner R, Arora R (2017) Insect pests and crop losses. Breed Insect Resistant Crops Sustain Agric 45–66

Smriti Sharma is a researcher affiliated with Punjab Agricultural University who co-authored the chapter "Insect Pests and Crop Losses" in the 2017 volume *Breeding Insect Resistant Crops* for Sustainable Agriculture In her work alongside Rubaljot Kooner and Ramesh Arora, she analyzes the impact of arthropod pests, estimating global crop losses of around 18–20% annually totaling over US $470 billion with disproportionately higher losses in the developing regions of Asia and Africa. Sharma emphasizes the urgent need for precise quantification of pest-related losses at every stage from field to consumption as a foundation for effective and sustainable pest management strategies. Her contribution offers critical insight into the scale of the problem and underscores the role of host-plant resistance and integrated pest management approaches in ensuring future food security.

7.Arif MS, Mukheimer A, Asif D (2023) Enhancing the early detection of chronic kidney disease: a robust machine learning model. Big Data Cogn Comput 7(3):144

Muhammad Shoaib Arif is a mathematics and data science academic affiliated with Prince Sultan University and Air University. Alongside co-authors Aiman Mukheimer and Daniyal Asif, he published the 2023 study "Enhancing the Early Detection of Chronic Kidney Disease: A Robust Machine Learning Model," in Big Data and Cognitive Computing. Their research introduced an innovative pipeline using iterative imputation, sequential scaling, Boruta feature selection, and K-Nearest Neighbors with hyperparameter tuning to detect CKD with perfect performance (100% accuracy, precision, recall, and F1-score) on the UCI dataset Arif's contributions highlight the powerful role of advanced preprocessing and ML techniques in healthcare, particularly for improving early CKD diagnosis and clinical decision-making

8.Asif D, Bibi M, Arif MS, Mukheimer A (2023) Enhancing heart disease prediction through ensemble learning tech niques with hyperparameter optimization. Algorithms 16(6):308

Daniyal Asif is a data science researcher at COMSATS University Islamabad, with a focus on machine learning applications in healthcare. In the 2023 study "Enhancing Heart Disease Prediction through Ensemble Learning Techniques with Hyperparameter Optimization," he led the development of an advanced predictive model combining Extra Trees, Random Forest, XGBoost, and CatBoost algorithms. Utilizing rigorous preprocessing, grid search hyperparameter tuning, and ensemble strategies on merged Kaggle datasets, his model achieved an impressive 98.15% accuracy for heart disease detection.

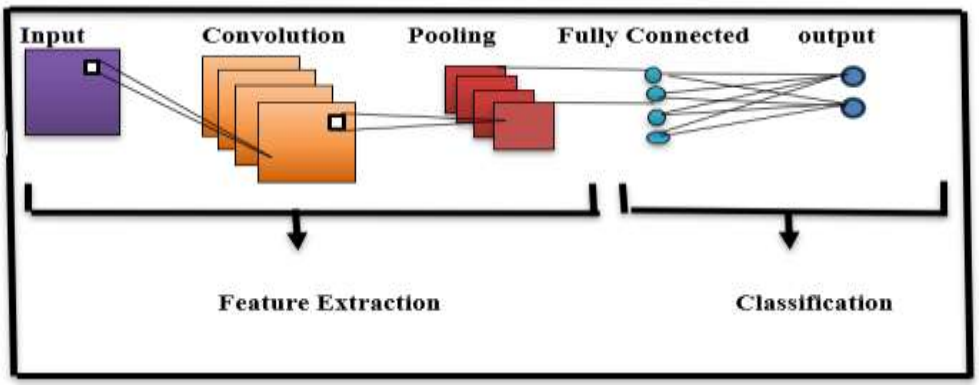
III. METHODOLOGY

This system uses a deep learning method to detect plant diseases, following steps like gathering images, preparing the data, training the model, and making predictions in real time. Convolutional Neural Networks (CNNs) work very well for analyzing pictures of plant leaves because they can learn important visual features on their own, without needing human help. Compared to other models like Random Forest, RNN, or LSTM, CNNs are better suited for handling image data, which makes them great at spotting small signs of disease in leaves. This can be leads to more accurate and reliable results.

3.1 CNN MODEL (CONVOLUTIONAL NEURAL NETWORKS)

The methodology of this project involves building a Convolutional Neural Network (CNN)-based deep learning system to detect and classify plant leaf diseases using image data. The overall process is divided into multiple stages: dataset preparation, image preprocessing, label encoding, model design, training and validation, evaluation, saving the model, and real-time prediction. Each of these steps is crucial to ensuring high model accuracy and practical usability. This makes them highly effective for recognizing disease patterns in plant leaves and typically results in very high classification accuracy

Fig 3.1.1 Architecture of CNN



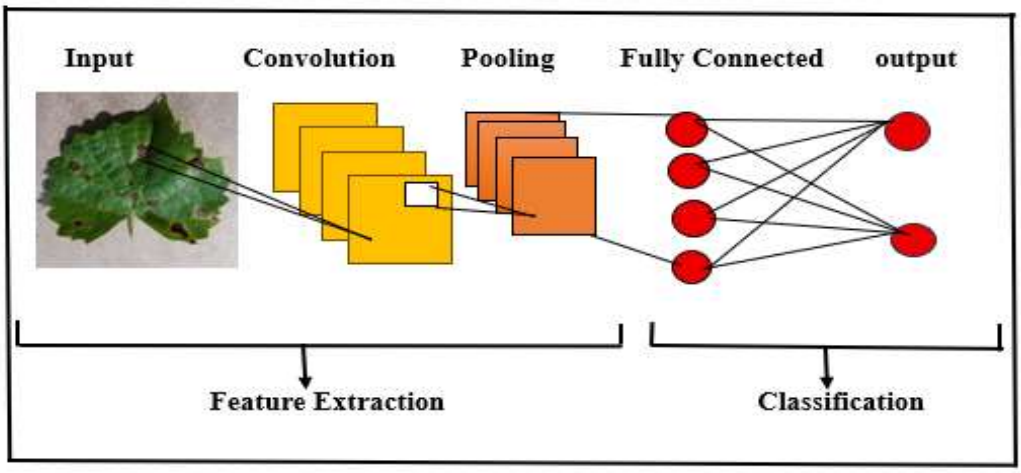
Dataset Collection and Preparation

The dataset used in this project is a subset of the publicly available PlantDisease dataset. For this study, images from the following four classes were selected: **Black Rot, ESCA, Healthy,Leaf Blight**

Image preprocessing

Image preprocessing plays a critical role in ensuring the dataset is clean, consistent, and ready for model training. It involves removing unwanted noise, enhancing relevant visual features, and standardizing the data. In this study, all images are resized to a uniform resolution of 224 x 224 pixels with 3 color channels (RGB) to ensure compatibility with the model and to accelerate training. Preprocessing also includes adjusting color, contrast, or texture and eliminating artifacts that could negatively affect model performance.

Fig 3.1.2 image preprocessing



Label Encoding and Dataset Partitioning

To make the class labels compatible with the deep learning model, they are transformed from string format to numerical values using Scikit-learn's. This conversion is crucial, as neural networks require inputs in numerical form.

After encoding, the dataset is divided into training and testing subsets in an **80:20 ratio**. This setup ensures that the model has access to a sufficient volume of data for learning while maintaining a separate portion to objectively assess its accuracy and performance.

Custom CNN Model Design

The model is constructed using TensorFlow and Keras, featuring a tailored CNN architecture specifically optimized for plant leaf classification. It comprises three convolutional layers, each followed by max pooling to progressively reduce spatial dimensions and extract key features.

The flattening layer transforms the 2D feature maps into a 1D vector, which is then passed through dense (fully connected) layers. The final output layer uses the softmax activation function to predict class probabilities across the four plant disease categories.

Model Structure:

- Conv2D(32) + MaxPooling2D
- Conv2D(64) + MaxPooling2D

- Conv2D(128) + MaxPooling2D
- Flatten + Dense(128) + Dense(4, softmax)

This structure is capable of learning and generalizing important features from leaf images

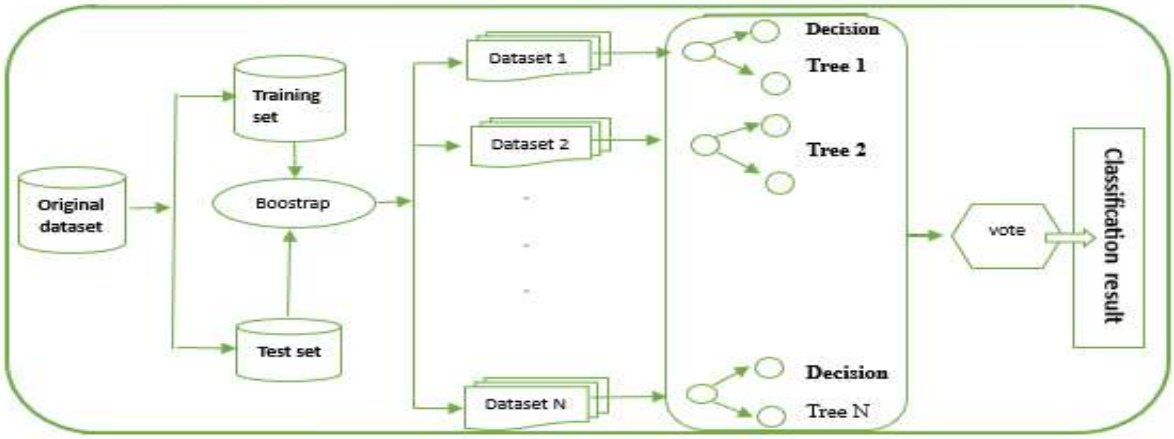
Model Compilation and Training

To train the CNN model, the Adam optimizer is employed due to its efficiency in adjusting learning rates dynamically. The loss function used is sparse categorical crossentropy, which is ideal for multi-class classification problems involving integer-labeled classes. The training process runs for 20 epochs with a batch size of 32. Validation data is included to evaluate the model's performance throughout training and to help mitigate the risk of overfitting.

3.2 RF MODEL(RANDOM FOREST)

Random Forest is an ensemble machine learning algorithm that operates by constructing multiple decision trees during training and outputs the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. It reduces overfitting and improves generalization by averaging the results of several de-correlated trees. Each tree is trained on a random subset of the dataset with replacement (bootstrap sampling), and features are randomly selected at each split to ensure diversity. It is particularly effective in handling high-dimensional data and noisy datasets. In plant disease classification tasks, Random Forest offers strong baseline performance due to its robustness and interpretability

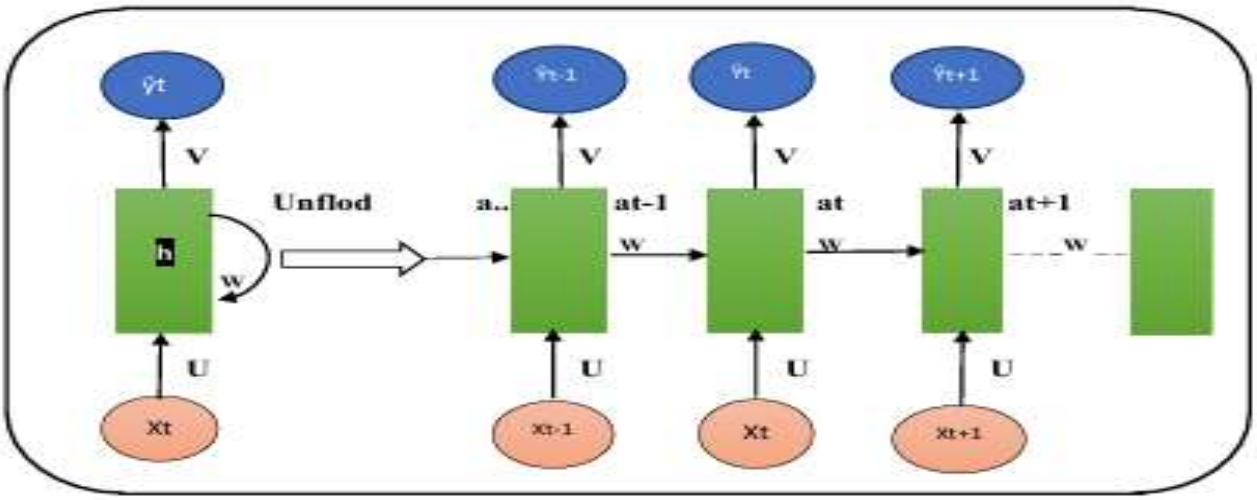
Fig 3.2.1 Architecture of RF



3.3 RNN Model (Recurrent Neural Networks)

Recurrent Neural Networks (RNNs) are a class of deep learning architectures tailored for handling sequential or time-dependent data. Unlike feedforward neural networks, RNNs maintain a form of memory through recurrent connections, allowing information from earlier inputs in a sequence to influence later outputs. In the context of plant disease detection using images, the image data can be reshaped into sequences, enabling the RNN to learn and interpret spatial patterns in a step-wise fashion. Though RNNs are typically used for applications like speech recognition or text analysis, they can also be adapted for agricultural use cases. Their strength lies in modeling temporal relationships, making them useful for sequence-based classification tasks when combined with appropriate preprocessing techniques.

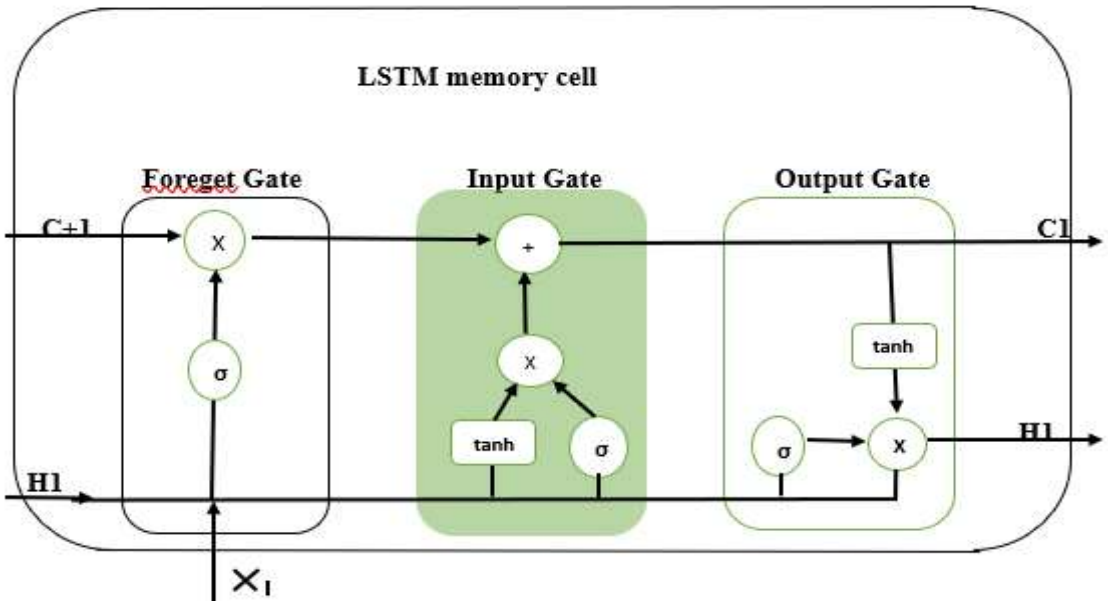
Fig 3.3.1 Architecture of RNN



3.4 LSTM Model (Long Short-Term Memory)

Long Short-Term Memory (LSTM) networks, a specialized form of RNNs, are capable of learning long-range dependencies within sequential data. In the context of plant leaf disease classification, the workflow begins with data collection and image preprocessing to ensure uniformity in input dimensions. For LSTM implementation, image data is reshaped into sequential formats suitable for temporal analysis. The model architecture typically includes LSTM layers followed by dense layers and a final SoftMax output layer for multi-class classification. Performance is thoroughly evaluated using metrics such as accuracy, precision, recall, F1-score, and a confusion matrix. The model delivering the best results is used for predictions, where processed test images are input to the network, and the predicted disease label is displayed alongside the image. This structured approach enables efficient and comparative analysis of LSTM's effectiveness in detecting plant diseases.

Fig 3.4.1 Architecture of LSTM



IV.RESULT:

After training, the model is evaluated using the test dataset. It achieved a high test accuracy of 98.97%, indicating that the model is capable of accurately classifying plant diseases on unseen data. The Convolutional Neural Network (CNN) model trained on the PlantDisease dataset, consisting of four disease achieved a high classification accuracy of approximately 98.97% on the test set. The preprocessing pipeline included resizing images to 128x128 pixels, converting grayscale images to RGB, and normalizing the dataset for consistency across different plant disease categories. During testing, the trained model successfully predicted unseen image data, providing strong evidence of its ability to generalize well to real-world images of grapevine leaves. The system also incorporates a simple image classification interface where users can input an image path, and the model will output the predicted class with high accuracy

Comparison table

Model	Acuracy	Precision	Recall	F1-Score
CNN	**0.98**	0.99	0.99	0.99
RF	**0.84**	0.87	0.85	0.86
RNN	**0.64**	0.82	0.80	0.81
LSTM	**0.69**	0.89	0.88	0.88

V. CONCLUSION

In this work, a deep learning model based on Convolutional Neural Networks (CNN) was effectively designed and implemented to classify plant leaf diseases. By applying systematic preprocessing techniques like image resizing and normalization, the input data was standardized, enhancing training stability and performance. The model demonstrated exceptional accuracy, confirming CNN's suitability for visual classification tasks in agricultural settings. This work lays the foundation for smart agricultural tools that can assist farmers with early disease detection, reducing crop loss and improving yield. The developed system efficiently differentiates between healthy leaves and four distinct disease types through visual symptoms. The entire pipeline—from image collection and preprocessing to training, augmentation, and fine-tuning—was methodically followed and explained. The model achieved a per-class precision ranging from 91% to 98%, with an overall classification accuracy of 98.97%, highlighting its reliability. With further

optimization, this system can be integrated into mobile or cloud platforms to provide real-time support for farmers and agricultural experts in the field.

VI. ACKNOWLEDGMENT

I would like to express my heartfelt gratitude to my guide, Asst Prof. **Ijjini Grace Asha Roy**, for her unwavering support, valuable guidance, and constant encouragement throughout the duration of this project. Her insights and mentorship were instrumental in shaping the direction and success of this work. I also extend my sincere thanks to the faculty members of the Department of Information Technology & Computer Applications, Andhra University College of Engineering, for providing the infrastructure, resources, and a conducive learning environment that made this project possible.

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